Grignard Reagent: Synthesis of an Alcohol

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Chem 2081-003

Introduction

In this experiment, a Grignard reagent was formed using magnesium, anhydrous ether and bromobenzene to produce phenyl-magnesium bromide. A mixture of acetone and ether was then reacted with the phenyl-magnesium bromide to form 2-phenyl-2-propanol, a tertiary alcohol. Ammonium chloride was then added to separate the 2-phenyl-propanol from waste upon the completion of day one. The product's highest stable boiling point was taken to confirm identity during day two.

Materials and Methods

The first day begins by plugging in your sand bath and setting it to 60. A rubber septum is cleaned for the reaction tube. The reaction tube is sealed immediately after removal from the open flame. Approximately 0.050 g of magnesium is placed in the reaction tube. Next 0.50 mL of anhydrous ether is added on top of the magnesium. The glass rod is used to swirl the pieces for about 30 seconds. Then 0.50 mL of anhydrous ether is placed in the 10 mL Erlenmeyer flask and 0.660 mL of the bromobenzene is added. Some of the ether/R-Br is drawn using a needle and syringe from the Erlenmeyer flask and about 24 drops are added to the reaction tube. The pink needle is used to ensure venting. The contents of the reaction mixture is mixed by agitating the bottom of the reaction tube. After the reaction has started, the rest of the ether/R-Br mixture is added dropwise over 10-15 minutes. The rate of addition is controlled. The beaker containing the mixture is warmed in a beaker of warm water for 10-15 minutes. Once the reaction is no longer vigorous, the setup is left in the warm water for 10 minutes. If the volume of the solution is less than 1 mL, anhydrous ether is added to reach 1 mL. A solution of 0.600 mL acetone in 0.50 mL of anhydrous ether. The ketone solution is added to the Grignard reagent dropwise and continuously over 10-15 minutes to maintain a gentle reflux. The tube is placed in a beaker of warm water for 10 minutes to complete the reaction. If the mixture is not at 2.5 mL add technical

ether to increase the volume past 2.0 mL. After the reaction is complete, the reaction tube is placed in an ice bath. Approximately 1 mL of saturated ammonium chloride is added dropwise. While the ammonium chloride is being added, the spatula is twirls to agitate the mixture. A pipette is used to move the organic product to a clean reaction tube. The product is washed using the pipette method with about 1.0-1.5 mL of saturated NaCl solution. Once the layers separate, the organic layer is moved to a clean reaction tube. The organic solution is treated with solid anhydrous sodium sulfate. The solution is capped with a good cork and kept until Day 2 of the experiment. Day 2 of the experiment begins by removing the sample from the drying agent and placing it in a clean reaction tube with a boiling chip. A simple distillation apparatus is assembled by attaching the distillation head directly to the reaction tube. The apparatus is wrapped in glass wool from the level of the liquid to the arm of the distillation head. The sand bath is raised until the sample begins to boil. The product is distilled, and the boiling point ranges are recorded. The distillation is stopped when the temperature drops or when only a small amount of sample remains in the tube. If 0.3 ml or less of sample is collected, then the distillation will not be performed. Instead, the thermometer would be brought down in the setup to just above the sample. The highest stable temperature is recorded.

Results

Component	**Recorded Boiling Point Estimate**
Final Product Day 2	70 ° C

Discussions

In this experiment, A Grignard reagent was formed, phenyl-magnesium bromide, and a mixture of acetone was then reacted with the phenyl-magnesium bromide to form an alcohol.

Due to qualitative assessment, this does not necessarily match up with the results obtained, as the generally recorded boiling point of 2-phenyl-2-propanol is 202 °C. This can be explained by breaking down a couple of variables. There was an issue with activating the magnesium by simple abrasion. This could be helped by adding iodine which would have added in activating the magnesium. Another variable is the use of acetone. Acetone does not readily activate with Grignard reagents and Grignard reagents are highly sensitive to water. These are important to note when analyzing the product. The failure to fully react the intermediate Grignard reagent to a 3 ° alcohol could have been the case for the severe boiling point difference for the expected product. The temperature could have been off due to manual error having to do with manual error and not placing the thermometer close enough to the actual reaction to properly measure the temperature.

References

Keller L, Joseph K, and Miguel S. 2022 Fall. CHEM 2071 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University.

Experiment 3 - Grignard Reagent

Procedure

1) Plug in sand bath in 10 amp outlet → set power controller to 60

2) Take hand towel, clean dry rubber septum for reaction tube, + cork size #4 for 10mL Erlenmyer flask. Glassware w/ no paper label

3) Seal reaction tube + flask immediately after removing from heat Don't open until room temp. Open only to add/remove material.

4) Weigh + place into reaction tube 0.0486 (0.05g) of Mg.

5) Add 0.50 mL anhydrous ether on top of Mg. Use glass rod to mix ~ 30 sec.

6) Place 0.50mL anhydrous ether in 10mL Erlenmyer flask Add 0.660 mL of bromobenzene

7) using syringe w/needle, draw ether/R-Br mixture from 10 mL Erlenmyer flask + inject ~24 drops into reaction tube w/ Mg + ether — Recork flask

8) Use pink needle (inserted through septum) for venting

9) Mix contents of reaction by agitating the bottom

10) Continue adding drops (R-Br mix) → AFTER reaction is underway until R-Br gone — Be sure to keep mixing

12) When reaction slows down, place mixture in beaker w/ 30-35°C water for 5-10

error - any water → improve w/ glove box

13) While tube in water bath, prepare 0.600 mL acetone in 0.5 mL of anhydrous ether — same flask as ether/R-Br mixture

14) Add ketone solution to Grignard Reaction (glide) via drops for next 10-15 min to maintain gentle rate - Agitate

15) Place tube in 30-35°C beaker of water for 10 min

16) Volume should be ~2.5 mL; if not add technical ether

17) When reaction complete, place reaction tube in ice bath using 100 mL beaker. Add 1 mL of NH_4Cl via drops + stir

18) Pipette product to clean reaction tube

19) Using pipette, wash product w/ 1.0-1.5 mL of saturated NaCl. Allow layers to seperate + transfer organic layer to clean reaction tube

20) Treat organic solution w/ solid anhydrous sodium sulfate

21) Cap sample w/ "good cork"

Experiment 3 Day 2

1. remove sample from drying agent, put in clean rxn tube w/ bc
2. if >.3 mL then distill, assemble simple distillation by attaching distillation head directly to rxn tube
3. use collection vial to collect product & make sure arm goes into vial
4. wrap apparatus in glass wool from level of liquid in rxn tube to distillation head, don't cover arm of distillation
5. raise ~~set~~ sand bath until sample boils, distill & record the boiling ranges
6. stir sand while distilling, record temp @ 1st & last drop.
7. stop distillation when thermometer drops ~~&~~ or when only a small amount ~~&~~ remains
8. if <.3 mL don't distill, bring thermometer to right above sample
9. bring sand bath up & record highest stable temperature measured, use this value to identify unknown
10. set power back to 0 & turn off power, turn off sand bath & clean all glassware

start @ 20° → ethanol

got to 110° ← not H_2O

Lab Performance

Exp 3 D1

Name: Abhi *Section:* *Room:* *Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves.

○ ○ ○ ○ ✗ (5pts)

Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks.

○ ○ ○ ○ ✓ (5pts)

Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts).

○(1pt) ○(1pt) ○(1pt) ○ (1pt) ✓ (5pts)

Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts).

○(1pt) ○ (5pts) ✓ (5pts)

Working independently or equal partner participation during the entire lab.

○ ○ ○ ○ ✓ (5pts)

Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace.

○ ○ ○ ○ ✓ (5pts)

Maintain clean and organized personal space.

○ ○ ○ ○ ✓ (5pts)

GTA notes:

GTA signature: *Final Performance Grade:* 2

Lab Performance

Exp 3 D2

Name: Abbi Starr *Section:* 003 *Room:* *Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves.	○ ○ ○ ○ ○ (5pts)
Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks.	○ ○ ○ ○ ○ (5pts)
Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts).	○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts)
Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts).	○(1pt) ○ (5pts) ○ (5pts)
Working independently or equal partner participation during the entire lab.	○ ○ ○ ○ ○ (5pts)
Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace.	○ ○ ○ ○ ○ (5pts)
Maintain clean and organized personal space.	○ ○ ○ ○ ○ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* [ ]